

# Technology, Social Construction of

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The social construction of technology (SCOT) is one approach among several constructivist ways of studying science and technology that emerged in the 1980s. Here, “constructivist” means that the truth of scientific facts and the working of technical artifacts are studied as accomplishments; that is, as being constructed, rather than as intrinsic properties of those facts and machines (→ Constructivism).

The phrase *social construction* was first used by Berger and Luckmann (1966) in their “treatise in the sociology of knowledge.” Building on the phenomenological tradition, and particularly on the work of Alfred Schutz, they argue that reality is socially constructed and that these processes of social construction should be the object of the sociology of knowledge. Berger and Luckmann are concerned with the reality of social institutions and their focus is on society at large, rather than on sub-cultures such as science and technology. Then, scholarship developed around such themes as the social construction of mental illness, deviance, gender, law, and class. In the 1970s the social construction of scientific facts followed (Latour & Woolgar 1986/1986), and in the 1980s the social construction of artifacts (Bijker et al. 1987).

Constructivist studies of science and technology come in a variety of forms, mild and radical (Sismondo 1993). The mild versions merely stress the importance of including the social *context* when describing the development of science and technology; for example Douglas’s (1987) history of radio broadcasting. The radical versions argue that the *content* of science and technology is socially constructed. In other words, the truth of scientific statements and the technical working of machines are not derived from nature but are constituted in social processes (→ Communication Theory and Philosophy; Objectivity in Science).

## THE ORIGIN AND DEVELOPMENT OF THE SOCIAL CONSTRUCTION OF TECHNOLOGY

An important, though negative, starting point for the social construction of technology was to criticize *technological determinism*, which was taken to comprise two elements: (1) technology develops autonomously, and (2) technology determines to an important degree societal development. This view was seen as an intellectually weak research strategy because it entails a teleological, linear, and one-dimensional view of technological development. It was considered politically debilitating because technological

*The International Encyclopedia of Communication*, First Edition. Edited by Wolfgang Donsbach.

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DOI: 10.1002/9781405186407.wbiect025

determinism suggests that social and political interventions in the course of technology are impossible, thus making a political debate on technology a futile endeavor. To bolster this critique, it was necessary to show that the working of technology was socially constructed – with the emphasis on *social*. The unit of analysis was the single artifact. This was a choice for the “hardest possible case.” To show that even the working of a bicycle or a lamp was socially constructed seemed a harder task, and thus – when successful – more convincing, than to argue that technology at a higher level of aggregation was socially shaped.

The agenda of demonstrating the social construction of artifacts by detailed analysis at the micro level resulted in a wealth of *case studies*. A few years later, the program was broadened in two ways (Bijker & Law 1992). First, questions were raised at meso and macro levels of aggregation as well. Second, the agenda was broadened to include the issue of technology’s impact on society, which had been bracketed for the sake of fighting technological determinism. Concepts developed for this agenda were “technological frame,” and various conceptualizations of the obduracy of technology (→ Framing Effects). The unit of analysis was broadened from the singular technical artifact to the more comprehensive and heterogeneous socio-technical ensemble. The emphasis now was on *construction* rather than on *social*.

Current research in SCOT combines empirical case studies with more general questions about the modernization of society, politicization of technological culture, and management of innovation. It has become increasingly difficult (and unfruitful) to observe the boundaries between the various approaches within the broader social construction of technology: research collaboration and conceptual combinations have emerged between, for example, the actor-network approach, the social construction of technology (in the narrow sense; SCOT), and gender and technology studies (→ Gender and Discourse). Connections are developing with studies in the social sciences more generally, including politics, economics, and law.

## SCOT AS A HEURISTIC FOR RESEARCH

As a heuristic for studying technology in society, the social construction of technology can be laid out in three consecutive research steps (Bijker 1995). *Key concepts* in the first step are “relevant social group” and “interpretive flexibility.” An artifact is described through the eyes of relevant social groups. Social groups are relevant for describing an artifact, when they explicitly attribute a meaning to that artifact. Thus, *relevant social groups* can be identified by looking for actors who refer to the artifact in the same way. For describing the high-wheeled *ordinary* bicycle in the 1870s such groups were, for example, bicycle producers, young athletic *ordinary* users, women cyclists, and anti-cyclists. Because the description of an artifact through the eyes of different relevant social groups produces different descriptions – and thus different artifacts – the researcher is able to demonstrate the “interpretive flexibility” of the artifact. There is not one artifact, but many. In the case of the *ordinary* bicycle, there was the Unsafe machine (through the eyes of women) and there was the Macho machine (through the eyes of young male *ordinary* users). For women the bicycle was a machine in which your skirt

got entangled and from which you frequently made a steep fall; for the “young men of means and nerve” riding the bicycle was a means to impress people (including young ladies!).

In the second step, the researcher follows how *interpretive flexibility* diminishes, because some artifacts gain dominance over the others and meanings converge – and in the end, one artifact results from this process of social construction. Here, the key concept is “stabilization.” It describes the result of the process of social construction, which can take several years while the degree of stabilization slowly increases.

In the third step, the processes of stabilization described in the second step are analyzed and explained by *interpreting* them in a broader theoretical framework: why does a social construction process follow this course, rather than that? The central concept here is “technological frame.” A technological frame structures the interactions among the members of a relevant social group, and shapes their thinking and acting. It is similar to Kuhn’s concept of → paradigm, with one important difference: “technological frame” is a concept to be applied to all kinds of relevant social groups, while “paradigm” was exclusively intended for scientific communities. A technological frame is built up when interaction around an artifact begins. In this way, existing practice guides future practice, though without logical determination. The cyclical movement thus becomes: artifact – technological frame – relevant social group – new artifact – new technological frame – new relevant social group, and so on. Typically, a person will be included in more than one social group, and thus also in more than one technological frame.

This three-step research process amounts to sociological analysis of an artifact to demonstrate its interpretive flexibility; description of the artifact’s social construction; and explanation of this construction process in terms of the technological frames of relevant social groups.

## SCOT AS A THEORY OF TECHNOLOGY IN SOCIETY

The social construction of technology also provides a theoretical perspective on the relation between technology and society, and on their joint development or co-evolution. The development of technology can be explained as a social process in which a variety of relevant social groups participate. This social process does not stop when an artifact leaves the factory, but continues when users give the technology its specific usage and meaning. Laws of physics and economics are not irrelevant, but insufficient to characterize technology’s development.

Analogously, the development of social institutions in modern society cannot be fully understood without taking into account the role of technology. The social fabric of society may be kept together by such institutions as church, capital, government, labor, communication, and education; but where would all these be without technology? Social order in modern society can only be explained by reference to technology (Latour 1992). This requires a view of technology’s impact on society. This can be done by conceptualizing the hardness or *obduracy of technologies*.

An artifact can be hard in two distinctly different ways. The first form, “closed-in hardness,” occurs when the humans involved have a high inclusion in the associated

technological frame. For example, students well versed in the use of mobile phones will react differently to a malfunctioning phone than the old-fashioned author of this article (→ Mobility, Technology for). Students may start modifying the network selection, switch the battery, or tinker with the preferences menu. Only after some time will it occur to them that they could also look for a landline phone: they were “closed-in” by the mobile phone technology. The author of this article, when confronted with a non-working cell phone, might try to ring a second time but then soon give up and look for a landline phone or write a letter. He can barely locate the power switch, and experiences the second kind of impact by technology: “closing-out obduracy.” He sees no alternative but to leave the technology aside and pick up his fountain pen. In both cases the technology has an impact on these people, but in completely different ways. These two forms of hardness of technology can also be seen on the societal level. The automobile technology, for example, exerts a “closed-in hardness” on the inhabitants of Los Angeles: much differentiation within auto culture, but no alternatives outside it. The standardization of mains power voltage and wall plugs mean a “closing-out obduracy” to most people: accept it by buying the right plug and apparatus, or do not use electricity (→ Communication Technology Standards).

## THE POLITICIZATION OF TECHNOLOGICAL CULTURE

We live in a technological culture: our modern, highly developed society cannot be fully understood without taking into account the role of science and technology. The social construction of technology offers a conceptual framework for discussing the democratization of this technological culture. The social construction of technology approach not only gives an affirmative answer to Winner’s (1980) question “Do artifacts have politics?” but it also offers a handle to analyze these politics. Technology is socially (and politically) constructed; society (including politics) is technically built; and technological culture consists of socio-technical ensembles (Bijker 2006).

Studies that elaborate this agenda do not exclusively extend the SCOT framework, but typically draw on social constructivist technology studies in the broad sense. One of the most fruitful bodies of work is the analysis of gender and technology (Wajcman 1991, 2004; Cockburn & Ormrod 1993). Another domain of research focuses on the politicization of → information society and information and communication technologies (see Schmidt and Werle [1998] for a fruitful approach via standardization issues).

The issue of political decision-making about technological projects acquires a special guise under the light of SCOT. If it is accepted that a variety of relevant social groups is involved in the social construction of technologies and that the construction process continues through all phases of an artifact’s life cycle, it makes sense to extend the set of groups involved in political deliberation about technological choices. Several countries are experimenting with consensus conferences, public debates, and citizens’ juries. One of the key issues is the role of expertise in public debates. The SCOT approach suggests that all relevant social groups have some form of expertise, but that no one form – for example, that of scientists or engineers – has a special and a priori superiority over the others (Bijker 2003, 2004).

SEE ALSO: → Communication Technology Standards → Communication Theory and Philosophy → Constructivism → Framing Effects → Gender and Discourse → Information Society → Mobility, Technology for → Objectivity in Science → Paradigm

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